Advanced active quenching circuits for single-photon avalanche photodiodes

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ABSTRACT

Commercial photon-counting modules, often based on actively quenched solid-state avalanche photodiode sensors, are used in wide variety of applications. Manufacturers characterize their detectors by specifying a small set of parameters, such as detection efficiency, dead time, dark counts rate, afterpulsing probability and single photon arrival time resolution (jitter), however they usually do not specify the conditions under which these parameters are constant or present a sufficient description. In this work, we present an in-depth analysis of the active quenching process and identify intrinsic limitations and engineering challenges. Based on that, we investigate the range of validity of the typical parameters used by two commercial detectors. We identify an additional set of imperfections that must be specified in order to sufficiently characterize the behavior of single-photon counting detectors in realistic applications. The additional imperfections include rate-dependence of the dead time, jitter, detection delay shift, and "twilighting." Also, the temporal distribution of afterpulsing and various artifacts of the electronics are important. We find that these additional non-ideal behaviors can lead to unexpected effects or strong deterioration of the system's performance. Specifically, we discuss implications of these new findings in a few applications in which single-photon detectors play a major role: the security of a quantum cryptographic protocol, the quality of single-photon-based random number generators and a few other applications. Finally, we describe an example of an optimized avalanche quenching circuit for a high-rate quantum key distribution system based on time-bin entangled photons.

Keywords: Single-photon detector, Avalanche photodiode, Photon detector imperfections, Quantum key distribution, Random number generation

1. INTRODUCTION

Detectors of light capable of detecting a single photon, so-called "single-photon detectors" or "single-photon counters," are used in large variety of scientific research areas and commercial applications, such as quantum information and quantum communication research, light radar (LIDAR), particle counting and sizing, gas analysis, time-resolved spectroscopy, nuclear and particle physics, astronomy, etc. In this paper, we restrict our attention to photon detectors based upon a specific type of semiconductor photon sensor, namely avalanche photodiodes (APDs) operating in Geiger mode. APDs capable of counting photons in the Geiger mode are usually named single-photon avalanche diodes (SPAD).

An ideal single-photon detector generates one logical pulse, for example TTL or NIM standard, for each photon that falls upon the sensitive area of the SPAD sensor. Of course, in reality, due to the limitations imposed by the physics of the SPAD and imperfections of the electronic circuits required to amplify and shape the signal, there are various deviations from this ideal picture. Manufacturers typically characterize their single-photon detectors by specifying a small set of parameters, typically the detection efficiency, the dead time and the dark counts. Less often, they further specify the afterpulsing probability and the single photon arrival time resolution (jitter). However, they almost never specify conditions under which these parameters are valid or make a sufficient description of the detector.

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In this work, we present an in-depth analysis of the active quenching process and identify intrinsic limitations and engineering challenges of single-photon detectors based on SPADs. Based on this analysis, we investigate the range of validity of parameters that are typically used to characterize commercial single-photon detectors and identify an additional set of imperfections that should be specified to sufficiently characterize the behavior of single-photon counting detectors in realistic applications. These additional imperfections include detection rate dependence of the dead time, rate dependence of the photon arrival time resolution (jitter), detection delay shift, twilighting, temporal distribution of afterpulsing, and artifacts of the electronics. We find that those behaviors can lead to unexpected effects or strong deterioration of the detector's performance with respect to what would be expected from the reduced set only. Specifically, we discuss implications of these new findings on the security of quantum cryptographic protocols and the quality of single-photon-based random number generators as examples of applications in which single-photon detectors play a major role. Finally, we present an avalanche quenching circuit with improved characteristics optimized for the quantum key distribution system based on time-bin entangled photons.

2. SPAD STRUCTURE AND ACTIVE QUENCHING

As an example of a typical SPAD structure, Fig. 1 illustrates a cross section of the commercial sensor SAP500 manufactured by Laser Components GmbH. We use this sensor in our detector described in Sec. 4. It is a reach-through structure illuminated from the back side, sensitive in visible (VIS) and near-infrared (NIR) wavelength spectral regions [1]. A photon entering through the bottom side may convert into a single free charge carrier in the conversion region (π-region), which is situated between a p+ layer and a p+ region. The efficiency of conversion is termed "quantum efficiency" (QE). The conversion region in this particular SPAD is only about 25 µm thick, which is on the thinner side for the reach-through structure. However, a special characteristic of the SAP500 is that parts of the bottom and top surfaces are covered with a metal layer that acts as a mirror and effectively increases the length of the available photon-conversion path. This improves the QE in the long-wavelength end of the sensitive spectrum, while at the same time keeping a good photon arrival timing performance (jitter). A broad-band anti-reflective (AR) coating on the entrance surface minimizes reflectance for the incoming photons. Yet another benefit of the short conversion region is the possibility of operating at a low breakdown voltage (about 125 V at 22 ^{o}C, while the other SPADs with similar spectral QE typically operate between 250 V and 500 V).

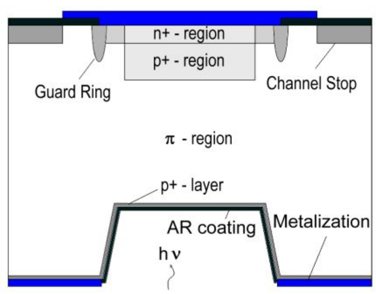

Figure 1. Cross section of the SPAD SAP500 manufactured by Laser Components GmbH. The active area is 0.5 mm in diameter. The photon conversion region is situated between the p^+ layer and p^+ region and is only 25 µm thick. The bottom and the contact part of the top side are covered by metalized layers whose purpose is to enhance containment of a photon and thus its conversion to a free carrier.

In a SPAD sensor biased above the Geiger breakdown voltage, a single-photon detection starts by converting the photon into a single charge carrier in the low-doped conversion region (π - region). In this region the electric field is low, enabling the charge carrier to quickly (in about 10 ps) reach a saturated velocity and to drift into the PN region, typically in about ~100 ps. In the PN region, consisting of p+ and n+ regions, the electric field is high enough to enable the carrier to acquire enough energy to generate a self-sustaining cascade (avalanche).

While the efficiency of photon conversion can reach almost 100% in a certain SPAD-based wavelength region, the probability of avalanching depends strongly on the bias voltage above the Geiger threshold V_{BR}, the so-called "excess voltage" V_E. The avalanching probability is zero when $V_E \leq 0$ and it rises with $V_E > 0$ to a certain maximum value which, together with other inefficiencies, defines the overall detection efficiency of a SPAD. Since the dark counts also rises with V_E, there is a practical limit or an optimum value for the excess voltage, which depends on the intended

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application. The non-unity detection efficiency is the most common imperfection of SPAD-based single-photon detectors.

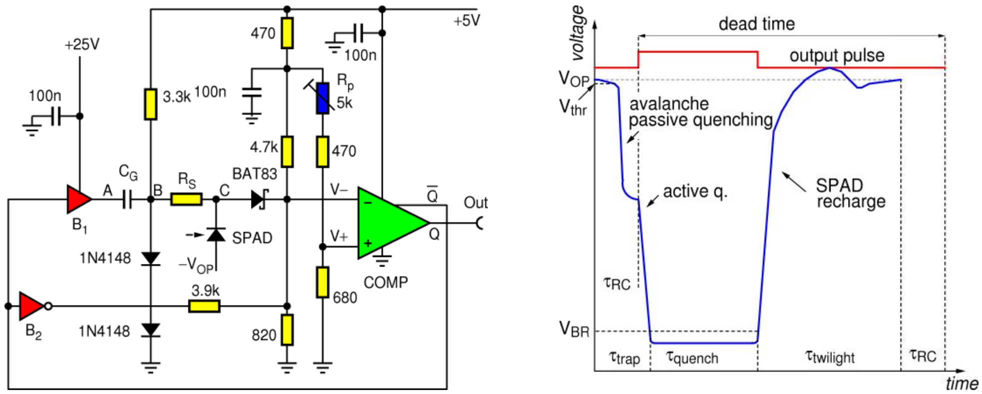

Figure 2. A realistic active quenching circuit with the double feedback loop (left). Timing analysis of the AQ circuit (right). Voltage across SPAD during a quenching sequence (blue, color online), where: τ_{trap} time period during which deep states (traps) are filled, τ_{quench} = time period during which SPAD is biased below the Geiger threshold, τ_{twilight} = time zone in which SPAD is (partially) sensitive to single photons but amplifying stage is shut down, τ_{RC} = propagation delay of the quenching stage, V_{OP} = nominal operating voltage of the SPAD, V_{thr} = pulse discriminating level, V_{BR} = Geiger breakdown voltage.

Once started, an avalanche can last for a relatively long time (miliseconds) and it must be quenched before the next photon can be detected. Quenching can be done by quickly lowering the bias voltage below the Geiger threshold and restoring it after a predetermined hold-off time. This is best done by help of active electronics components (e.g., transistors), which enable fast lowering and restoring of the bias. During an avalanche, the charge multiplication process results in the generation of a sizeable charge or current that is comfortably above the readout noise of the subsequent amplifying stage and thus a single photon can be reliably detected. A realistic active-quenching circuit with a double feedback is shown in Fig. 2 (left), previously described in [2]. In this topology, the SPAD is connected to the negative bias voltage $-V_{OP}$. The avalanche causes the comparator COMP to go high and trigger the two inverting buffers I_1 and I_2 to generate a large-amplitude quenching signal at point A after a total propagation delay T_1. This will lower the voltage at the point C via the capacitive coupling capacitor C_G and series resistance R_S, thus bringing the SPAD's bias below the Geiger threshold. In the process described in [2] the quenching cycle will result in the SPAD bias cycle illustrated in Fig 2 (right). We see that there is a minimum time span between any two output pulses: the "dead time", yet another well-known detector imperfection.

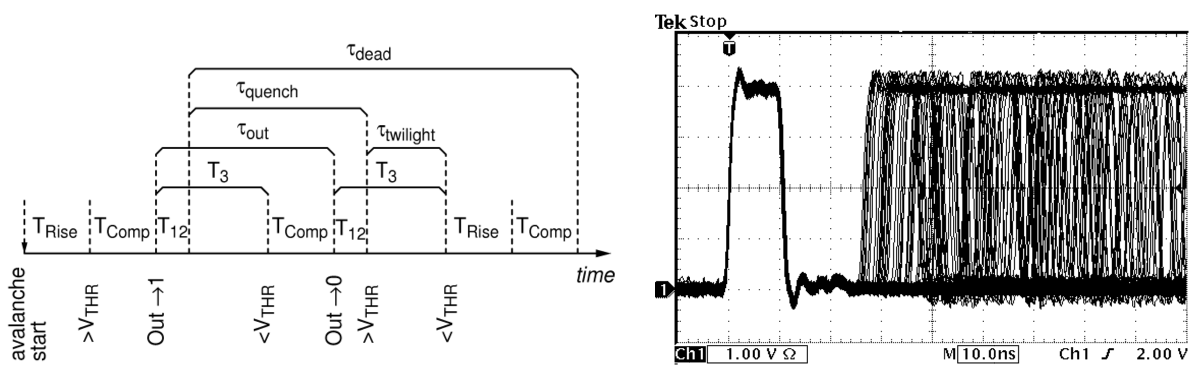

Figure 3. Timing diagram of a quenching cycle (left), see the text. Cumulative output of a single-photon detector illuminated by Poissonian light from a LED diode attenuated to yield approximately 50,000 counts per second on average (right). The dead time is defined as minimum time delay between the trigger pulse (on the right) and the next pulse, and is about 26 ns for this detector.

The dead time of a single-photon detector can be visualized by a cumulative display of detection pulse pairs appearing at the output of the detector, as shown in Fig. 3 (right). The dead time, defined as the minimum time between two consecutive pulses, is widely taken to be equal to the photon pair resolution time. There is perhaps a historical reason for

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this belief, because the pulse pair resolution is identical to the dead time for photomultiplier detectors, but the same does not hold for SPAD-based photon counters. Namely, restoring the bias voltage across the SPAD after quenching can only be done at a finite rate, and may feature a few ripples before settling at the nominal value (see Fig. 2).

Therefore, in order to avoid retriggering, a second loop going through I_3 is engineered such that it restores the sensitivity of the comparator COMP only after the settling has been completed. The part of the dead time between the end of quench and start of sensitivity of the comparator COMP, denoted by τ_{twilight} on Fig 2. (right), is called the "twilight zone" [3] because, during that time, the bias is partially restored and the SPAD can go into avalanche even though the detector is technically in the dead time.

In the practical circuit shown in Fig. 2, the quenching pulse starts a few nanoseconds after the dead time, has a flat-bottom that lasts for about 10.5 ns, and ends with the twilight zone of about 3 nanoseconds. An avalanche occurring in the twilight zone eventually produces an output pulse, but it appears shifted to *after* the dead time. Because the avalanche may happen sooner or later during the twilight, and since the recovery time of the electronics may be affected by it, twilight events may appear in a short time period (a few nanoseconds typically) after the dead time. It is important to note that twilighting is an unavoidable consequence of the free-running quenching loop process (i.e., not specific to this particular circuit) and therefore it cannot be avoided. Twilighting is thus yet another imperfection of SPAD-based photon detectors, but unlike previously described ones it is never mentioned nor characterized in the datasheets of commercial detectors. We will analyze this effect in more detail in the next section.

The delays within the cycle are described in detail in the timing diagram shown in Fig. 3 (left). Measured values of timing parameters are: the time from the avalanche till comparator is triggered $T_{\text{rise}} = 2$ ns, the comparator propagation delay $T_{\text{COMP}} = 4.5$ ns, delay of buffers I_1 and I_2 $T_{12} = 3$ ns, delay of the buffer I_3 $T_3 = 6$ ns. From these values, we can calculate the twilight zone $T_{\text{twilight}} = 3$ ns, the output pulse width $T_{\text{out}} = 10.5$ ns and dead time $T_{\text{dead}} = 26$ ns. The latter two parameters can be verified by the direct measurement shown in Fig 3 (right).

As shown in [4], the avalanche should be quenched as soon as possible to minimize the effect of afterpulsing; also, the avalanche current should be limited, which is partially done here by the resistor R_S. Afterpulsing in a silicon (Si) SPAD is usually well explained by deep trap states having a unique lifetime, which leads to a decaying exponential probability distribution function (p.d.f) of afterpulses. Also, Si SPADs with two or more distinct lifetimes have been observed [5], [6]. In InGaAs/InP SPADs, in which lifetimes of the deep trap states form a continuum such that the p.d.f. of afterpulses resembles a power-law [7], although no such result has been reported for a Si SPAD. Thus, afterpulsing is yet another common imperfection in SPAD-based photon counters.

SPAD detectors in the VIS-NIR range suffer from small to intermediate level of false counts that appear when the detector is kept in complete darkness, so called "dark counts". These can be lowered by cooling the SPAD, but this may elevate afterpulsing [8], [9].

Finally, photon arrival time is measured with an uncertainty, usually named "jitter," that has two contributions: one comes from the variation of the drift time from the place of photon conversion to the avalanche region, and the other is associated with statistical fluctuation of the avalanche current [10]. Jitter is an unavoidable imperfection found in all SPAD-based photon counters.

The main technical challenges in virtually any active avalanche quenching design are: (1) engineering the propagation delay time between avalanche start and its quenching to be as short as possible in order to minimize afterpulsing; (2) keeping the twilight zone as short as possible; (3) achieving short dead time; and (4) minimizing the contribution of the electronics to the overall detector jitter.

3. HIDDEN IMPERFECTIONS IN SINGLE-PHOTON DETECTORS

The "standard" imperfections described above, namely non-unity detection efficiency, dead time, afterpulsing, jitter and dark counts, are the most common parameters used for characterization of single-photon detectors. However we find that they do not sufficiently characterize the detection processes and that other parameters exist that must be taken into account. We have already met one: twilighting.

To study twilighting, we start by measuring a number of time intervals between two consecutive output pulses when a detector is subjected to a weak CW laser light. We use a well-known photon-counting module SPCM AQRH-12 (SN22097) from Excelitas, Canada (formerly Perkin Elmer). We have already seen in the previous section (Fig. 3

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(right)), one can visualize the dead time by visualizing a series of detection events on an oscilloscope. However, in this test, we measure the time intervals by means of a high-resolution time-to-amplitude converter (Ortec 567) and create a histogram shown in Fig. 4. Three types of events are clearly visible: detection of real photons, which have been removed from the plot, afterpulses, and an excess of events appearing in a short period just after the dead time. The excess corresponds to the events (avalanches) that start in the twilight zone and whose detection is shifted to after the dead time. Also shown is a single-exponential fit of afterpulses, yielding a single exponential lifetime of 33 ns and afterpulsing probability of 2.38%. The shortest observed time interval (28.5 ns) corresponds to the dead time of the particular detector used for tests presented in this section.

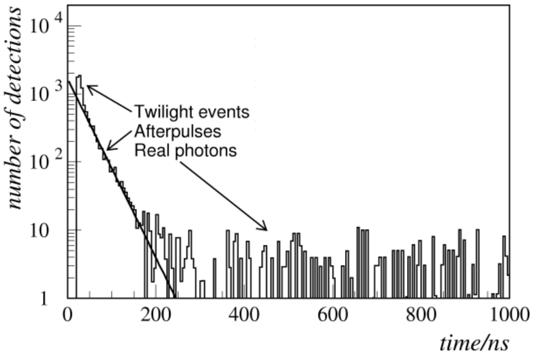

Figure 4. Histogram of time intervals between subsequent pulses recorded from the detector Excelitas SPCM AQRH-12. The detector is illuminated with a CW laser light of $\lambda = 635$ nm attenuated such that the average frequency of detections is about 50 kcps. Three types of events are clearly visible: detection of real photons, afterpulses and an excess of events just after the dead time, due to twilight events. Also shown is an exponential fit of afterpulses. The shortest time interval (of 28.5 ns) is due to the dead time of the detector.

We also find that the "standard" parameters themselves are not independent of measurement conditions. We have again used the SPCM AQRH-12 detector. In this and subsequent tests, a picoecond pulsed laser (pulse duration 220 ps FWHM, wavelength $\lambda = 635$ nm) is fired periodically at a rate of 10 MHz. The laser is coupled to the detector via a variable coupler set to achieve a detection rate of (from left to right): 50 kcps, 200 kcps, 500 kcps, and 1 Mcps. We note that the photon arrival-time resolution worsens from the initial 430 ps at 50 kcps to 1035 ps at 1 Mcps. At the same time, the mean value of the photon detection delay slides by 610 ps towards longer times. In case of a varying detection rate, this additional shift would further smear the time resolution.

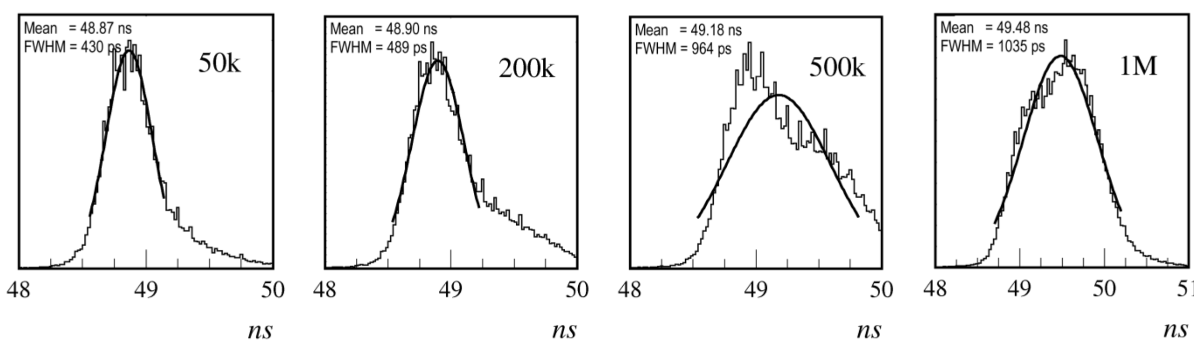

Figure 5. Histogram (a linear scale in arbitrary units) of jitter of the Excelitas SPCM AQRH-12 photon counting module as a function of the mean detection rate. A picoecond pulsed laser was fired periodically at a rate of 10 MHz and coupling to detector was set such as to achieve the detection rate of (from left to right): 50 kcps, 200 kcps, 500 kcps, and 1Mcps. We see that the resolution worsens from initial 430 ps to 1035 ps and that at the same time the mean value slides by 610 ps towards longer times.

Next, we probe the photon detection in the "twilight" zone. To perform this measurement, a pair of weak consecutive laser pulses separated in time by ΔT is sent to the detector. Each pulse has a probability of about 1/20 to cause a detection so that each pulse wavepacket has essentially a single-photon or no photon. The detection efficiency of the first photon in a pair and the second photon in a pair when the first photon has been detected have been measured and the ratio of second and first photon efficiencies is plotted as a function of the delay ΔT. Two commercial detectors were

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tested: the Excelitas SPCM AQRH-12 discussed above (Fig. 6, left) and Micro Photon Devices SPD-050 [11] (Fig. 6, right). The dead time of each detector is shown as a dashed line on the respective plot. From this, one clearly sees that photons can indeed be detected during the dead time with essentially the nominal efficiency. However, the output pulse corresponding to the second photon cannot appear before the dead time. This means that photons in the twilight zone get detected with a delay not smaller than the dead time minus the time when the photon caused the avalanche in the twilight zone. For example, photons occurring at the beginning of the twilight zone will appear *shifted* in time by at least as much as the duration of the twilight zone. This could be a considerable problem in applications sensitive to photon timing in which photons can appear in the twilight zone of a detector.

To study the shift, we measure the detection time of the second photon in events where both photons were detected and plot the difference between the times when the photon was detected and when it was expected to be detected. The result of the test for the SPCM AQRH-12 detector is shown in Fig. 7 (left). As expected, in the twilight zone, the detection shift is a linear function of the photon pair delay, but more intriguing is that the shift is apparent even some 15 ns after the dead time. It finally falls below 100 ps for photon pair delay $\Delta T > 47$ ns.

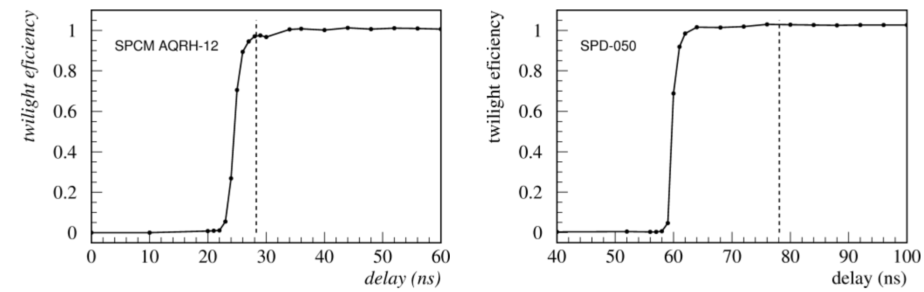

Figure 6. Twilighting in the SPCM AQRH-12 (left) and SPD-050 (right) detectors. Shown is the detection efficiency of the second photon in a pair relative to the efficiency of the first photon when the first photon is detected. The horizontal axis is the delay between the two photons. The dead times of 28.5 ns and 78 ns, respectively, are indicated by the dashed line in the respective graph.

There is yet another interesting question: How is the time resolution (jitter) affected by the temporal separation between two subsequent detections, notably by the twilight zone and the dead time? To answer this question, we use the detector SPCM AQRH-12 and illuminate it with photon pairs as explained above. We consider only those events in which both photons in a pair are detected and measure the timing jitter of the second, as a function of the delay ΔT between the two photons. The results are shown in Fig. 7 (right).

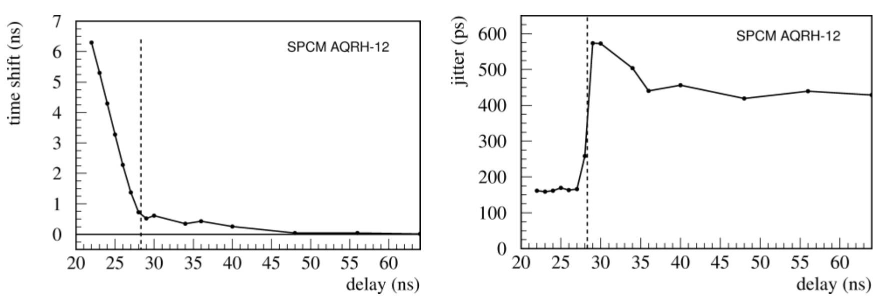

Figure 7. Time shift between the true and measured photon arrival time for the second photon in a pair (if both photons have been detected), as a function of the time interval between the two incoming photons (left). Time resolution (jitter) of the second photon in a pair if both photons have been detected (right).

The average detection frequency is set low, about 50 kcps per photon, so the jitter should correspond to the best value recorded in Fig. 5 (about 430 ps FWHM). We see that the photons appearing up to 10-15 ns after the dead time suffer from an enlarged jitter that asymptotically approaches the nominal jitter for large ΔT. This is probably due to an incomplete recovery of the bias voltage across the SPAD, which would also explain the observed time shift. On the other hand, photons detected in the twilight zone appear to have smaller jitter but this is an illusion because the uncertainty of their arrival extends over the whole twilight zone and the observed small jitter corresponds only to the delayed detection pulse.

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The SPD-050 is a detector made with special care to achieve the best timing resolution in the visible range, (typically between 35 ps and 50 ps FWHM), and very low afterpulsing, (typically < 0.5%) [11]. For both of these goals, a prolonged quenching period, long enough dead time, and sufficient time for bias settling seem necessary. As a result, the SPD-050 does not suffer the jitter and shift variations when ΔT is greater than the dead time, but it does feature an enlarged twilight zone.

4. AN IMPROVED CUSTOM-BUILT DETECTOR

While effects of the above-described imperfections may be small or avoidable for some applications, for others they may become prohibitive. For the purpose of performing experiments with quantum key distribution (QKD) protocols (that will be described in Sect. 5), we have undertaken an effort to build an improved single-photon detector based upon a novel active quenching circuit. Our QKD protocol requires a detector optimized for: (1) high detection efficiency; (2) short dead time; (3) low jitter; (4) reduced twilighting; and (5) high stability of the jitter and detection delay shift with respect to the detection frequency of up to 4 Mcps. Afterpulsing probability up to $\sim$3% and dark counts up to a couple kcps are acceptable. We identified the red-optimized SAP500-T8 from Laser Components as the best candidate for the purpose. This SPAD contains within the package a double-stage thermoelectric cooler (TEC) that can be conveniently used to stabilize the junction temperature and contribute to the stability of the detector's performance. Its peak detection efficiency can reach 73% at our target operating wavelength of 710 nm.

A schematic diagram of our novel custom-built avalanche quenching (AQ) circuit is shown in Fig. 8. The SPAD is held at a constant temperature of -10 ^{o}C, found to be an optimum tradeoff between afterpulsing probability and dark count rate. The positive high voltage (HV) supplies the bias voltage 15 V above the Geiger threshold. It is a circuit similar to the one shown in Fig. 2, with the main difference that the capacitive coupling of the quenching signal to the SPAD is now replaced with galvanic coupling. Namely, the capacitive coupling has a problem that, as the average detection frequency goes up, the coupling capacitor C_G has less time to completely charge and discharge, which leads to the lower quenching step and reduction of the direct current base voltage at point B and consequently the discriminating threshold for the avalanche sensing. These effects lead to elongation of the dead time, higher jitter, and detection delay shift. The last two imperfections are also seen in the SPCM AQRH-12 in Fig. 5. To minimize these unwanted effects, the capacitive coupling is replaced by a galvanic one in the new circuit. Additionally, the return-path impedances and circuit poles have been carefully engineered to minimize the settling time of the quenching signal, which shrinks the twilight zone by an appropriate choice of the propagation delay time of the buffer DLY achieved using a single non-inverting logic AND gate (Fig. 8). The quenching signal at the point A has a peak-to-peak amplitude of about 25 V, enough to comfortably quench the SAP500-T8 operating at excess voltages up to 20 V. However, we selected an excess voltage of 15 V and an operating temperature of -10 ^{o}C to optimize the trade-off between the dark counts rate, detection efficiency, afterpulsing probability, and jitter for the applications described below.

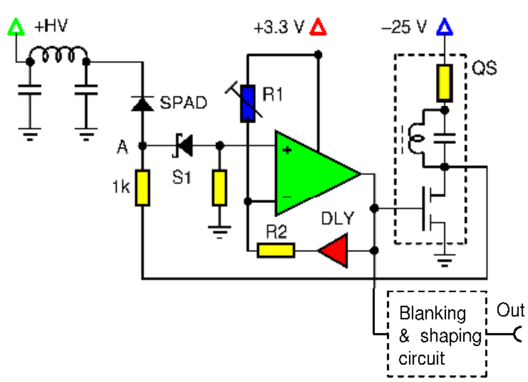

Figure 8. Schematic diagram of the improved avalanche quenching circuit. The SPAD SAP500-T8, made by Laser Components GmbH, is held at a constant temperature of -10 ^{o}C. The positive high voltage HV supplies the bias voltage 15 V above the Geiger threshold.

Here are some of the performance parameters we measured with the custom-built detector made by the use of the above described AQ circuit (Fig. 8) and the SAP500-T8 operated at 15 V excess voltage and at -10 ^{o}C: detection efficiency $= (73 \pm 2)\%$ at 710 nm, dead time $= 24$ ns, afterpulsing probability = 3.2%, and jitter = 155 ps FWHM at low detection rates (< 100kcps). A test of stability of jitter versus detection rate, identical as performed for the SPCM

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AQRH-12 in Fig. 5 (laser pulsed at 10 MHz), shows that the jitter steadily rises from 155 ps at 50 kcps to 161 ps at 1 Mcps detection rate, while at the same time the detection delay stays within 20 ps. When tested at more demanding conditions (laser pulsed at 33.3 MHz) the jitter varies as follows: 156 ps for detection rates of up to 100 kcps, 164 ps at 1 Mcps, and 185 ps at 4 Mcps. When subject to the test of the jitter stability versus time separation of pair of detected photons, whose results for the Excelitas detector are shown in Fig. 7 (right), the custom-built detector shows better stability, as shown in Fig. 9 (middle).

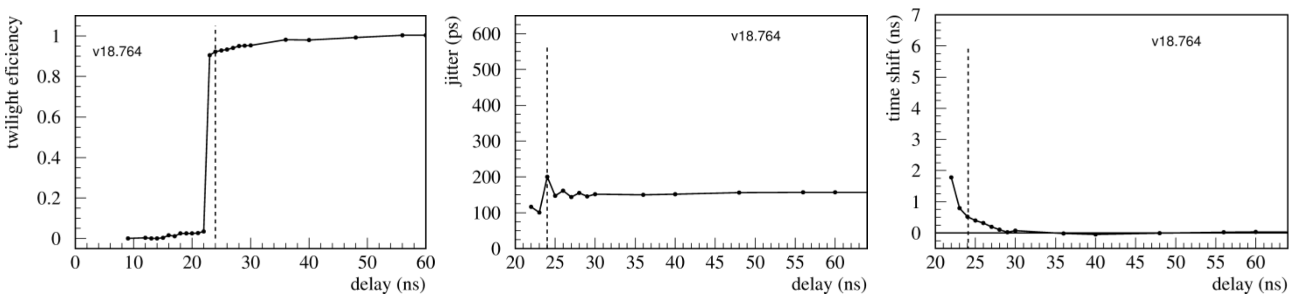

Figure 9. Study of the custom-built detector. Shown are the following parameters of photon detection as a function of time delay from the last detected photon: twilighting (left); jitter (middle) and the detection time shift (right). The dead time of 24 ns is indicated by the dashed line.

Still, some variation of the dead time with the detection rate and some twilighting is observed. We measured an elongation of the dead time by about 2.5 ns when the detection rate of photons emanating from the CW laser source was varied from zero to 30 Mcps by means of adjusting the light intensity. In order to eliminate any dead time drift as well as to cut off any twilight events that appear right after the dead time, the output of the detector is filtered through a digital "blanking circuit" whose task is to block any pulse that is closer than a predetermined time period $\tau_D > \tau_{\text{dead}}$ to the previous pulse. In this way τ_D becomes the new dead time of the detector. The blanking circuit is effective in eliminating the dead-time fluctuations and tightening the twilight zone to only about 1.5 ns, as shown in Fig. 9 (left). The circuit also shapes the output pulse such that it has a constant shape. Fast restoring of the bias after quenching enabled good stability of jitter versus the second photon delay, as shown in Fig 9 (middle). Apparently, the most sensitive parameter to bias restoration is the photon detection delay, shown in Fig. 9 (right); it falls below 100 ps only for detections about 28 ns or more apart from the previous detection.

5. APPLICATIONS OF SINGLE-PHOTON DETECTORS

In this section we present two applications of single-photon detectors in which different sets of detector imperfections play a significant role.

Our first example is a simple quantum random number generator (QRNG) based on the well-known beam-splitter principle [12], [13], schematically shown in Fig. 10. Attenuated CW laser light enters the polarizing beamsplitter PBS, causing an equal average rate of photon detections in single-photon detectors D0 and D1. For simplicity we assume that the two detectors have the same characteristics. Let us see what happens if the detectors are plagued with afterpulsing and long dead time, the two most common imperfections found in SPAD-based photon detectors. It has been noted [14] that, at a low rate, afterpulsing will have a major influence causing a positive serial autocorrelation among generated bits, while, at the high detection rate, the dead time will cause large negative serial autocorrelation.

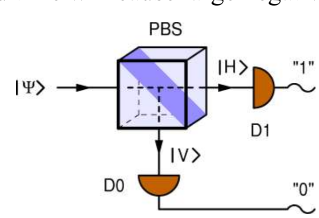

Figure 10. Beam-splitter-based random number generator. Intensity of the incoming continuous light wave is divided in two equal parts by the beamsplitter PBS, causing an equal average rate of photon detections in detectors D0 and D1, which signify generatuon of bit values 0 and 1, respectively. Since the time order of detections from either detector is random, so is the sequence of generated bits.

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Namely, at a vanishingly low detection rate, when a detector has just detected a photon, the most probable event is an afterpulse in the same detector, that is, the generation of the same bit value which leads to positive serial autocorrelation among bits. In the absence of dark counts, the serial autocorrelation coefficient is equal to the afterpulsing probability. This causes a large error even with only moderate afterpulsing probability. On the other hand, at a high detection rate, a detector that has made the last detection will be dead for some time, so the most probable event is a detection by the other detector, which causes a strong negative serial autocorrelation. This effect limits the highest attainable bit generation rate. In this process the twilighting as well as detection time shift have virtually no effect.

For this special application, there are ways of mitigating the negative effects of detector imperfections. Since detection time and detection location of the photons are independent, it is possible to improve the randomness of the system by combining the temporal and spatial information [15]. Alternatively, by applying a pulsed light instead of CW, it is possible to eliminate the problem of the dead time and significantly suppress afterpulses at the same time [16].

Our second example is a super-fast quantum key distribution (QKD) setup. In our protocol [17], two parties, Alice and Bob, share pairs of hyper-entangled photons [18] generated by spontaneous parametric downconversion (SPDC) in a pair of nonlinear optical BiBO crystals. Photons in a pair are simultaneously entangled in polarization, spatial mode, and time-bin degrees of freedom (DOF). Each DOF plays a different role in the overall QKD protocol: most of the raw key bits are encoded in the photon timing (which "pulse position modulation" allows up to 10 bits per photon to be shared), polarization entanglement is used to check for eavesdropping, and the spatial modes realize independent quantum communication channels whose purpose is enhancing the secret key rate. The experimental realization of one spatial DOF channel is shown in Fig. 11.

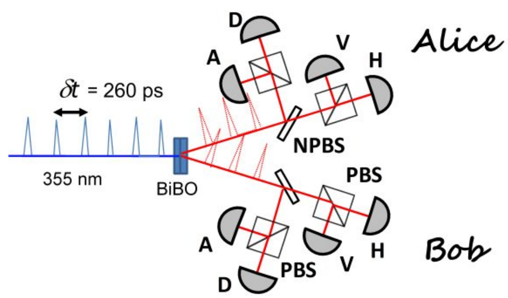

Figure 11. Experimental setup for our QKD system for one spatial mode. The non-polarizing beam splitter (NPBS) in Alice and Bob's setup randomly direct the photonic states to either the Horizontal (H)/Vertical (V) basis or the Diagonal (D)/Anti-Diagonal (A) polarization bases, where single-photons are detected and their time of arrival recorded.

A mode-locked diagonally (D) polarized pump laser fires light pulses of duration 5 ps FWHM and wavelength 355 nm, periodically with a period of 260 ps, thus generating a maximally entangled polarization state. The intensity of the laser pumping the downconversion crystals is adjusted so that Alice and Bob each receive on average 1 photon in a time "frame" consisting of 1024 successive time bins each 260 ps wide. This system allows, in principle, transfer of up to 10 bits per photon.

First, we perform a standard polarization check for QKD on the polarization side. We then assume an eavesdropper has no quantum non-demolition (QND) measurement that can measure timing without altering the polarization, and thus we use the bit error rate (BER) of the polarization degree of freedom to estimate the information leaked in timing as well. However, there is a simple attack Eve can do that doesn't alter the polarization but still measures timing: she blocks random time bins, thus enhancing the probability that Alice and Bob receive bits in a reduced set of bins. To detect this type of an attack, we produce decoy pulses at $\bar{N} = 0.1$ and $\bar{N} = 0.01$ by producing difference frequency generation at the BiBO crystal between a supercontinuum source and the pump laser, where $\bar{N}$ is the mean photon number of the decoy states. By measuring photon statistics, we can estimate how much blocking Eve is doing and correct for it in the privacy amplification stage.

There are several detector-sensitive aspects of this protocol. First, photons are incoming in narrow time slots (bins) that are only 260 ps wide. In order to obtain high entanglement purity in the time-bin degree-of-freedom, each detector must have jitter and detection delay shifts (when combined) significantly smaller than the bin width. Second, since the time bins are much shorter than the twilight zone of a detector, some of the photons will necessarily end up there. Such a photon, if detected, will suffer a shift of at least a few nanoseconds, which means that it will appear in a completely

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wrong time bin, directly contributing to the bit error rate (BER). Since the detection rate is high (about 1 Mcps per detector), a substantial portion of the photons (a few percent) will hit the detector during its dead time and immediately after it. Time shifts associated with twilighting and post dead-time recovery will thus play an important role in the BER and in limiting the maximum achievable secret key rate.

We obtain a raw key rates of around 2.6 MBit/s using the laser pump with the repetition rate of 1.925 GHz using the standard Excelitas SPCM AQRH detectors. At a higher repetition rate, the raw key rate declines because jitter of the detector is greater than the bin width. However, with our improved detectors, we can distinguish pulses at a rate of 3.85 GHz, allowing the system to achieve a raw key rate of 3.6 Mbit/s. The difference between the two detectors in the autocorrelation function $g^2(0)$ is illustrated in Fig. 12.

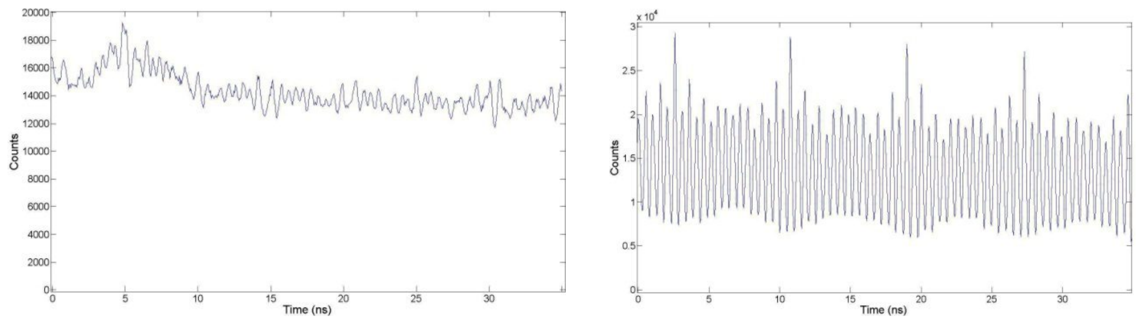

Figure 12. Autocorrelation plots for the Excelitas SPCM AQRH detector (left) and for the custom-made detector (right). These plots were made using a pulsed laser runs at 1.925 GHz, where we can still distinguish the pulses (despite the autocorrelation broadening) with both detectors.

Due to high rate of entangled photon pair detections by Alice and Bob, dark counts and (less so) afterpulsing are much suppressed as they contribute minimally to the signal-to-noise ratio (SNR) of the system and do not pose a problem, in contrast to the previous example of the beam splitter RNG.

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